AP PRECALCULUS · TOPIC 1.8

Rational Functions and Zeros

Section 1 — Multiple Choice
Circle the letter of the correct answer **SOLVED**

Name ..... Class _____ Date _____

No credit for unsupported answers in Section 2. Questions 1–8: 1 point each (8 points).

1 Let $r(x) = \dfrac{(x-3)(x+4)}{(x+1)(x-2)}$. What are all of the real zeros of r?

A. $-4,\ 3$ **B.** $-3,\ 4$

C. $-4,\ -1,\ 2,\ 3$ **D.** $-1,\ 2$

A. The numerator is zero at $x = 3$ and $x = -4$, and the denominator is not zero at either. B: sign error, reading each factor $x - a$ as $x = -a$. C: listed the zeros of the denominator as well. D: gave only the zeros of the denominator.

2 What are all of the real zeros of $f(x) = \dfrac{x^2 - 9}{x^2 - x - 6}$?

A. $-3,\ 3$ **B.** -3

C. 3 **D.** -2

B. $f(x) = \dfrac{(x-3)(x+3)}{(x-3)(x+2)}$. At $x = 3$ the denominator is 0, so 3 is not in the domain (hole). Only $x = -3$ remains. A: counted the hole $x = 3$ as a zero. C: kept only the excluded value. D: used the remaining denominator factor.

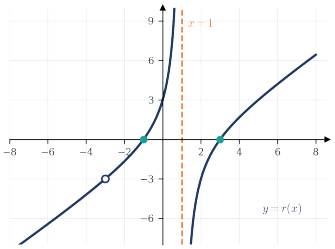
3 The graph of the rational function r is shown. It has x-intercepts at $x = -1$ and $x = 3$, a vertical asymptote at $x = 1$, and a hole at $(-3, -3)$. Which is the solution set of $r(x) \le 0$?

A. $(-\infty, -1] \cup (1, 3]$

B. $(-\infty, -3) \cup (-3, -1] \cup [1, 3]$

C. $(-\infty, -3) \cup (-3, -1] \cup (1, 3]$

D. $[-1, 1) \cup [3, \infty)$



C. The graph is on or below the x-axis for $x \le -1$ and for $1 < x \le 3$, but the hole at $x = -3$ is not in the domain. A: ignored the hole at $x = -3$. B: included the asymptote $x = 1$ as a closed endpoint. D: gave the intervals where $r(x) \ge 0$ instead.

4 Which list gives all of the real zeros of $g(x) = \dfrac{x^3 - x}{x^2 + 4}$?

A. $-1,\ 1$ **B.** There are no real zeros

C. $-2,\ 2$ **D.** $-1,\ 0,\ 1$

D. $g(x) = \dfrac{x(x-1)(x+1)}{x^2+4}$ and $x^2 + 4 \neq 0$ for every real x. A: divided out x and lost the zero at 0. B: thought a denominator with no real zeros means the function has no zeros. C: took the zeros of the denominator's factor $x^2 - 4$ by mistake.

5 What is the real zero of $r(x) = \dfrac{1}{x-2} - \dfrac{3}{x+4}$?

A. -7 **B.** 5

C. -5 **D.** r has no real zeros

B. $r(x) = \dfrac{(x+4) - 3(x-2)}{(x-2)(x+4)} = \dfrac{10 - 2x}{(x-2)(x+4)}$, zero at $x = 5$. A: set $x - 2 = 3(x + 4)$, cross-multiplying with the wrong sides. C: sign error in the numerator, solving $2x + 10 = 0$. D: subtracted the numerators $1 - 3$ only and concluded the numerator is never zero.

6 Which is the solution set of $\dfrac{x+2}{x-3} \le 0$?

A. $[-2, 3]$ **B.** $(-\infty, -2] \cup (3, \infty)$

C. $(-2, 3)$ **D.** $[-2, 3)$

D. Boundaries -2 (zero) and 3 (asymptote); the quotient is negative between them. A: included $x = 3$, where the denominator is zero. B: gave the intervals where the quotient is positive. C: dropped the zero $x = -2$ although $\le$ includes it.

7 Which function has exactly one real zero?

A. $f(x) = \dfrac{x^2 - 4}{x - 2}$

B. $f(x) = \dfrac{x^2 - 4}{x^2 + 1}$

C. $f(x) = \dfrac{(x - 2)(x + 1)}{x + 3}$

D. $f(x) = \dfrac{x^2 + 4}{x - 1}$

A. For the correct choice, $f(x) = x + 2$ for $x \neq 2$, so the only zero is -2 ($x = 2$ is a hole). B: has two zeros, ± 2, since $x^2 + 1$ is never 0. C: has two zeros, 2 and -1, neither excluded. D: has no real zeros, because $x^2 + 4 > 0$.

8 Let $f(x) = \dfrac{(x - 4)^2(x + 1)}{(x - 4)(x + 2)}$. Which statement is true?

A. f has zeros at $x = -1$ and $x = 4$

B. f has a zero at $x = 4$ only

C. f has exactly one zero, $x = -1$, and a hole at $(4, 0)$

D. f has zeros at $x = -2$ and $x = -1$

C. Cancel one factor $x - 4$: $f(x) = \dfrac{(x - 4)(x + 1)}{x + 2}$ for $x \neq 4$. f is undefined at 4, so 4 is not a zero, and the graph has a hole at $(4, 0)$. A: counted the hole on the x-axis as a zero. B: used only the squared factor. D: used the zero of the denominator $x = -2$ as a zero.

Rational Functions and Zeros

Section 2 — Free Response
Show all working; answers need units and limit notation where relevant

Question 9 Calculator allowed · 6 points

A biologist models the rate of change of the fish population in a lake, in hundreds of fish per month, by $R(t) = \dfrac{2(t-6)(t+1)}{t+3}$, where $t \geq 0$ is the time in months since the study began.

(a) Find the zero of R in the context of the problem. Explain why the other zero of the numerator is not used. DOK3 2 pt

$2(t-6)(t+1) = 0$ gives $t = 6$ and $t = -1$. The denominator is nonzero at both, but $t = -1$ is outside the domain $t \geq 0$. The zero is $t = 6$ months.

(b) Solve $R(t) \geq 0$ for $t \geq 0$ and interpret your answer in context. DOK3 2 pt

On $t \geq 0$ the denominator is positive and $t + 1 > 0$, so the sign of R is the sign of $t - 6$. $R(t) \geq 0$ for $t \geq 6$. After 6 months the population is not decreasing; before then it is decreasing (e.g. $R(0) = -4$).

(c) Find the time at which the rate of change is exactly 4 hundred fish per month. Give your answer to three decimal places. DOK3 2 pt

$2(t-6)(t+1) = 4(t+3) \Rightarrow t^2 - 7t - 12 = 0 \Rightarrow t = \dfrac{7+\sqrt{97}}{2} \approx 8.424$ months (the root $\dfrac{7-\sqrt{97}}{2} < 0$ is rejected).

Question 10 No calculator · 6 points

Let $r(x) = \dfrac{x^3 - 4x^2 - 5x}{x^2 - 2x - 15}$.

(a) Factor the numerator and the denominator, and state the real zeros of r. DOK3 2 pt

$r(x) = \dfrac{x(x-5)(x+1)}{(x-5)(x+3)}$. The numerator is zero at $0,\ 5,\ -1$, but $x = 5$ makes the denominator 0, so it is not in the domain. Zeros: $x = -1$ and $x = 0$.

(b) Find the coordinates of the hole in the graph of r. DOK3 2 pt

The cancelled factor is $x - 5$. The reduced function is $\dfrac{x(x+1)}{x+3}$; at $x = 5$: $\dfrac{5 \cdot 6}{8} = \dfrac{15}{4}$. Hole at $\left(5, \dfrac{15}{4}\right)$.

(c) Solve $r(x) \geq 0$. Write your answer in interval notation. DOK3 2 pt

Boundaries from $\dfrac{x(x+1)}{x+3}$: $-3,\ -1,\ 0$. Signs on $(-\infty,-3),\ (-3,-1),\ (-1,0),\ (0,\infty)$: $-,\ +,\ -,\ +$. Include the zeros -1 and 0, exclude -3 and the hole $x = 5$: $(-3,-1] \cup [0,5) \cup (5,\infty)$.

Answer key (Section 1): 1A 2B 3C 4D 5B 6D 7A 8C